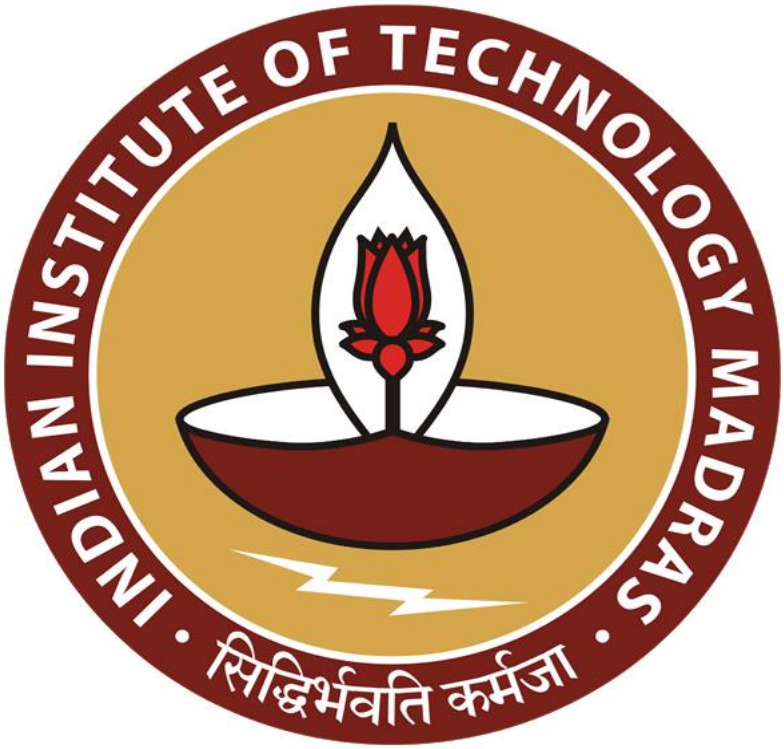


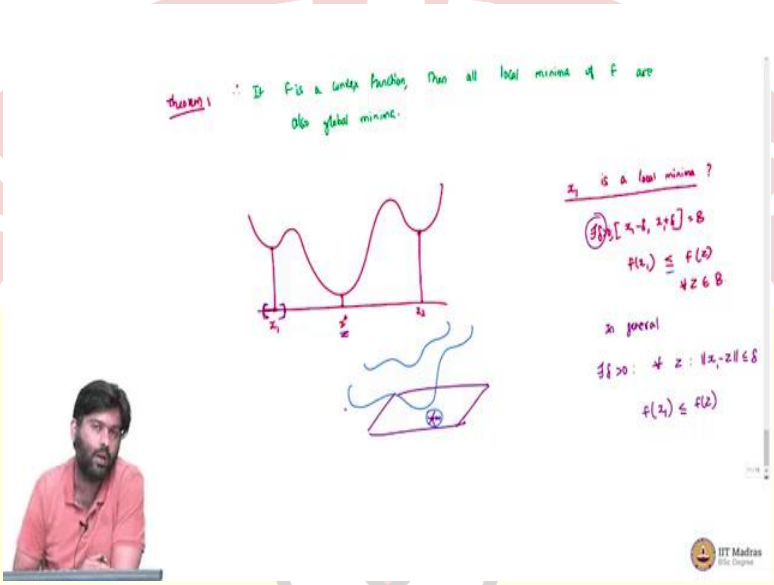

# IIT Madras

## ONLINE DEGREE

**Properties of Convex Functions**

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So, the next thing we are going to see and it is a nice fact which makes convex functions very, very interesting is that convex functions enjoy this nice property that, if f is a convex function then all local minima of f are also global minima. So, this is actually a theorem. So, let me remind you what a local minima is. So, we formally defined this. You might have functions like this and we call this point as a local minima. So, this is a local minima. Well, this is also a local minima, but also happens to be a global minima. Maybe here is another local minima. So, maybe this is x1, this is x2.

In general, let us call, we will, in this particular case, we are going to call x1 as a local minima. Why is it a local minima? So, minima, why is it local minima? The reason why is it a local minima, because there exists some interval around x1. So,me interval of length $\delta$ which means that you have, there exists $\delta$ which is positive such that $x - \delta$ and $x + \delta$ , if I look at this interval around x, my f of x around x1, so we are looking at f(x1) is less than or equal to f(z) for all z belonging to, if this call, let me call this set S or maybe we have used S already, so let me call the set B to indicate that some sense of a ball. Ball in one dimension is just an interval.

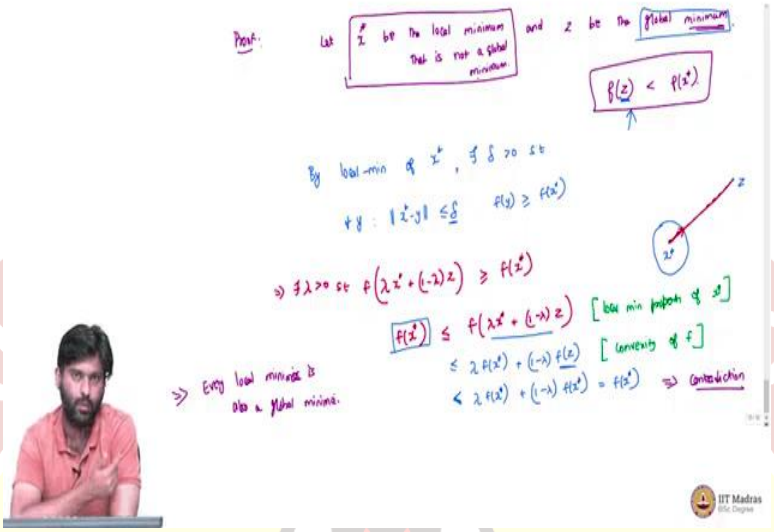
So, what are we saying? So, we are saying geometrically that if, there exists some interval around x1 so where if I look at the values of the function around in that interval x1 is the smallest. Now, in general, this is for higher dimension. This would mean that there exists some $\delta$ again greater than 0 such that for all z which is within the z, $\delta$ radius ball around x, so for all z such that the norm of x1 - z is less than or equal to $\delta$ . So, we have that f(x1) is less than or equal to f(z) . So, in two dimension, if you have a two dimensional function, so, and maybe this is a function which is like this.

Now, what this is saying is that I take some x1 and if it has to be a local minima, then I should be able to draw a small ball around it and with within that ball this function should have the least value at x1 or less than or equal to all other points within that ball. In particular, the interesting point to note that here is that, if this happens, then it means that I am allowed to move from x1 to any direction, because it is a ball. So, it is a complete ball, which means that I can move in any direction from x1. But if I do not move too much, so within $\delta$ , if I move in any direction, I will end up in a point where the function value is higher than where I am now. That is what a definition of a local minimum is.

Now, a global minimum is also a local minimum. It also has this definition. But then we say that it is not just for a small interval, but this holds for any $\delta$ , which means that you can go in any direction, however, how much however you want to, and you will still end up at a point which is less than the value of this evaluated at the global minimum. So, that is the only definition between local minimum and global minimum.

But as you can see, not all functions have global minimums as exactly, local minimums is exactly global minimums. I mean, there could be a point which is a local minimum, but not a global minimum. But the theorem says that this is not true for convex functions. So, if f is a convex function, then it necessarily has to be the case that if you have a local minima then it is also a global minimum. So, and this would be interesting. We will see the implications of this in a minute, but then let us try to understand why the statement is true next.

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So, let us try to prove this fact, so proof. Now, let us call this, let x, let me make it, maybe I will call it x* be the local minimum and z be the global minimum. And now we are saying that the x, the, not all global minimums are local minimums. That is what we are saying. So, which means that x* is a local minimum, but it is not a global minimum. What does that mean? That means that, well, the global minimum is strictly has value less than the local minimum. So, otherwise, these two guys would, x* would also be a local, global minimum.

So, now, this implies that immediately by our assumption that x* is a local minimum and z be a global minimum, let me make it more precise, let x*t be the local minimum that is not a global minimum. So, basically, we are trying to prove this by contradiction. So, we wanted to show that every local minimum is also a global minimum. Let us assume that there is an x* which is a local minimum, but not a global minimum. And now let z be indeed the global minimum.

So, which means that we know now that f(z) which is the smallest possible that you have among all possible sets has to be strictly less than f(x*). This has to necessarily happen. So, if it is not strictly less than, if it is even equal to, then it immediately goes against our assumption that x* is a local minimum that is not a global minimum. If it is equal then x* will also become a global minimum, so because its function value is as small as the global minimums one. So, we will assume that x* is not a global minimum, which means that x* is only locally minimum but then its value is greater than f(z).

Now, we know that by the definition of x* being local minimum, it necessarily has to have be the case, so by local minimum property of x*, there must exist some δ greater than 0 such that for all y in, for all y which is such x* - y is less than δ  f of y is actually greater than or equal to f(x*) . In that small epsilon or a δ ball around x* f(x*) is minimum. So, which means that everybody else is higher greater than or equal to f(x*) 's value within that small ball.

Now, the important point is that if I am at x*, then there is some small ball around me. I can move in whatever direction I want to move in this ball and I will still end up at a point which has a higher value or equal to my value. So, I would not go lesser than that, lesser than my current value will necessarily be greater than or equal to my value. Now, let us move in the direction which means that there has to be some direction which will take me along, if I go enough along the direction I will reach the global minimum, but I would not go as much as the global minimum, I will only move within my δ  ball in that direction.

So, this is x*, maybe this is z. I know that there is this ball around x*, where x* is, value is smaller than anybody else in this or less than or equal to anybody else in this ball. Now, let us say we move in this direction. So, we move in this direction. Now, if you, what do I mean by moving in this direction? Well, I can move this point such that I get a point which is some λ times x* + (1 - λz).

So, as I move λ, as I change λ, I can move from x* to z, from x* till z. But let me not go till z, I will only pick a λ, there should exist a λ, so which implies there must exist a λ greater than 0 such that the function value at this new point is greater than or equal to f(x*) . So, this has to happen. So, because I know that x* is local minimum this has to necessarily happen. In other words, let me write this the other so that will become easier. So, this is less than or equal to f of λ x* + 1 - λ z.

This has to necessarily happen. Why, because I know x* is locally minimum which means that there should be some amount that I can move in the direction towards z so that I can only increase the value. This should happen. Now, what do we know about the function? The function is a convex function. Because it is a convex function, we can do something now. We did the first step because of local minimum property of x*. Now, the next step we are going to do is by exploiting the fact that it is a convex function.

We know that if because f is convex, if you evaluate at any point between two points the function's value is only less than or equal to the original point, the interpolations. So, which means that this

will be less than λ times f(x*) + (1 - λ * f(z)) . This has to happen. Because why is this true, because of convexity of f. So, this is simply convexity of f.

Now, what is z? Z was not an arbitrary point. Z was indeed the global minimum. And because z is global minimum, we already know f(z) is less than f(x*) . So, because x* is a local minimum but not a global minimum, so f(z) is less than f(x*). Now, which means that this whole thing is strictly less than λ f(x*), the first time I am not doing anything, the 1 - λ also stays, but then f(z) I know is strictly less than f(x*). The strictness is important here. So, I will do this. But then what is this quantity. This quantity is simply λ f(x*) + 1 - λ f(x*). Now, no matter what λ you choose. This guy is simply going to be f(x*).

Now, what have we proved? We have just shown that you started with f(x*) and then we are showing that f(x*) is strictly less than f(x*) , but that is absurd. So, this is, this implies contradiction. This cannot happen. F(x*) cannot be less than f(x*) , which means that somewhere some assumption that we did must have been wrong and if you trace back the only assumption that we did was x* be a global, local minimum which is not a global minimum then necessarily that should have been wrong, so which means that every local minimum is also a global minimum.

So, now, the implication is that every local minimum, minima is also a global minima. So, this is a very interesting simple proof, but then a very far reaching, as far reaching consequences. Now, what does it say? It tells us that if you have a convex function and if you are trying to optimize that function and if you have a method that will find you local minimums then this theorem automatically tells you that it is, it suffices, because your local minimum will also be global minimum. So, which means that you would have actually found the global minimum of the function that you are trying to optimize by having a procedure which in general only produces local minimum.

Now, do we know of such a procedure? Yes, we know of such a procedure, which is our gradient descent algorithm. We argued that the gradient descent algorithm is guaranteed to produce local minimums. Now, we are saying that in the case of, if you have more assumptions about the function that the function itself is convex, then local minimums are also global minimas and so gradient descent is good enough for convex functions to produce optimal solutions. So, that would be the first main conclusion of using convexity.

In the future classes what we would see is we will try to see some examples of convex functions which are practically used in machine learning and we will see how we can, what kind of discussions that we, that will be important when we were, when you are trying to optimize these kind of functions. So, that would be what we will do next. With this I will stop this session. Thank you.

